Shayan Edalatmanesh, Ph.D.

Computational Physical Chemistry | DFT | ML Interatomic Potentials | Agentic AI | Scientific Software
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Professional Summary

Computational physical chemist (PhD 2023) with 7+ years of DFT and atomistic simulation of molecular, surface and low-dimensional systems. Experienced in building ML-accelerated and automated simulation workflows. Recent work combines **MACE ML interatomic potentials** with DFT, generating synthetic datasets for AI model training, and evaluating AI agents running ab initio calculations. Developer and maintainer of an extended version of **PPSTM**, an open-source STM simulation package in Python,part of the ProbeParticle Package. **20 peer-reviewed publications** (14 as lead theoretical contributor) in top journals, including the Nature family.

Experience

Postdoctoral Researcher Jan 2025 – Mar 2026
Spanish National Research Council (CSIC - CFM), San Sebastián, Spain
(with Prof. Nicolas Lorente)

- Used MACE ML interatomic potentials to pre-screen 40 adsorption geometries of a 90-atom **molecular switch** on Au(111) (570+ atoms, where brute-force DFT is prohibitive), narrowing to 12 candidates and cutting full VASP calculations by 70%. Manuscript in preparation (lead theoretical contributor)
- Built high-throughput DFT–STM simulation workflows for **transition-metal dichalcogenides** (WSe_2) with automated post-processing (DOS, charge analysis, STM visualisation), producing curated synthetic image datasets for AI model training and benchmarking
- Modelled π–d magnetism in metal–organic nanostructures with DFT & model Hamiltonians (*Angew. Chem.*, 2025)

Postdoctoral Researcher Jun 2023 – Dec 2024
Donostia International Physics Center (DIPC), San Sebastián, Spain
(with Prof. Thomas Frederiksen)

- Showed how substrate hybridization drives **topological phase transitions** and edge-state evolution in open quantum systems using model Hamiltonians (*Phys. Rev. B*, 2025)
- Studied STM-induced lifting as a platform for tuning non-Hermitian boundary states and substrate-adsorbate **charge transfer**
- Contributed charge-analysis functionality to *sisl*, an open-source tight-binding/DFT post-processing library

PhD, Theoretical Physical Chemistry Sep 2018 – May 2023
Palacký University & Institute of Physics of the Czech Academy of Sciences, Prague, Czech Republic
(with Prof. Pavel Jelínek)

- Simulated 2D hybrid perovskites, graphene nanoribbons, polymers and molecular motors for electronic and spintronic applications; studied topological phase transitions in 1D polyaromatic hydrocarbons & **graphene nanoribbons** with DFT (*Nature Nanotechnology*, *Nature Chemistry*, *Nature Communications*, *ACS Nano*)
- Extended the **Fireball DFT code** (Fortran) with Zak-phase calculations; wrote Python tools for real-space topological analysis, adopted by collaborators in surface science and tensor-network research
- Extended PPSTM for STM simulations of periodic materials (see Selected Projects)
- Co-supervised 1 MSc and 1 BSc student, whose work resulted in papers in *ACS Nano* and *Science Advances*

Visiting Researcher May 2022 – Jul 2022
Universidad Autónoma de Madrid, Spain (with Prof. José Ortega Mateo)

- Built DFT models to study molecular motors on metallic surfaces

Visiting Researcher Nov 2019 – Dec 2019
Aalto University, Finland (with Prof. Adam Foster)

- Developed a DFT-based dipole model to explain organic–inorganic 2D hybrid perovskites

Selected Projects

PPSTM: open-source STM simulation package in Python *Project info* · *GitHub*

- Extended to periodic systems via Bloch's theorem, enabling simulation of crystalline surfaces and slabs, not only isolated molecules
- Added constant-current scanning (3D grid + isosurface extraction) for experimentally comparable 2D/3D current maps, plus automatic structure overlay and standard file I/O
- Used by research groups internationally; citable software record on Zenodo

Agentic AI evaluation for ab initio simulation *Project info*

- Evaluated the Claude Science AI agent running VASP DFT calculations autonomously, scored on physical correctness, autonomy, error handling, reasoning and computational efficiency
- Prototyped automated scoring and a hybrid keyword + embedding retrieval pipeline to ground agent answers in VASP documentation

League of Legends match-outcome predictor *Project info* · *Notebook*

- Logistic regression in PyTorch on 2025 ranked-match data (40,410 player records)
- Full clean / train / tune / evaluate workflow (80/20 split)
- BCE loss, SGD, 1,000 full-batch epochs
- 75% test accuracy (74% with L2 regularization) & ROC AUC of 0.82

Publications, Service & Recognition

- **20 peer-reviewed publications** (1 first author, 3 co-first author, 2 corresponding author; lead theoretical contributor in 14) in *Nature Nanotechnology*, *Nature Chemistry* (×2), *Nature Communications* (×2), *Science Advances*, *ACS Nano*, *Advanced Materials*, *Angewandte Chemie*, among others; plus 1 arXiv preprint under review (with new experimental data).
  Full list: Google Scholar
- **Peer reviewer** for *Physical Review B* (4 manuscripts)
- **Invited chapter**, *Elsevier Encyclopedia of Interfacial Chemistry*: "Topological Quantum States in Carbon Nanostructures" (in preparation, 2026)
- **Fischer Scholarship**, *Outstanding International PhD Student of the Year*, awarded twice (Palacký University)

Technical Skills

Methods: DFT, tight-binding model Hamiltonians, molecular dynamics (i-PI), STM/AFM/KPFM simulations
DFT codes: VASP, FHI-aims, SIESTA, FIREBALL, Gaussian, ORCA
Technical specialties: MACE (MLIP), Magnetism, spin-orbit coupling, QM/MM, GW, topological band theory
Machine learning: Deep learning, regression analysis, data processing, predictive modeling, dataset generation
Programming: Python (PyTorch, NumPy, SciPy, ASE, Matplotlib, Pandas), Fortran, C++
Tools & HPC: Bash, git, LaTeX, ssh, SLURM, VESTA, VMD
Databases: Materials Project API, NOMAD

Education

PhD, Physical Chemistry 2018 – 2023
Palacký University & Institute of Physics of the Czech Academy of Sciences, Prague, Czech Republic
Thesis: Theoretical Study of Charge Transport in Nanostructures

MSc, Solid State Physics 2014 – 2017
Iran University of Science and Technology, Tehran, Iran
Thesis: The Effects of Monovacancies on the Transport Properties of Zigzag Graphene Nanoribbons

BSc, Physics 2008 – 2013
Iran University of Science and Technology, Tehran, Iran

Certificates

AI Agents and Agentic AI with Python & Generative AI (Vanderbilt University); Introduction to Neural Networks and PyTorch (IBM); Claude Code 101 (Claude Academy); Claude Code in Action (Claude Academy)

Languages & Interests

Languages: English (fluent), Persian (native), Spanish (B1), Czech (A2)
Interests: Football, photography, specialty coffee (former barista champion), cinema, math rock